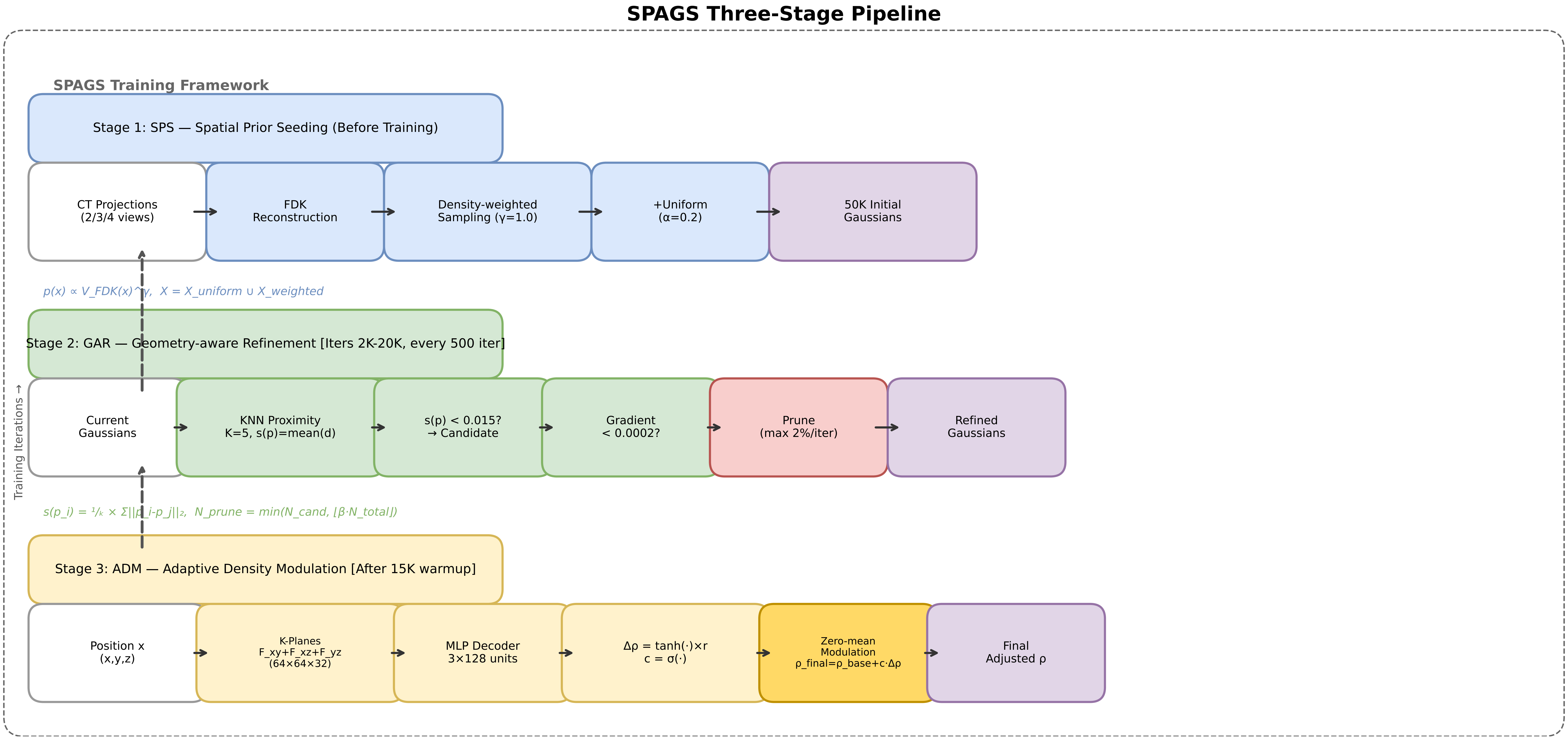
SPAGS Three-Stage Pipeline
SPAGS Training Framework
Stage 1: SPS — Spatial Prior Seeding (Before Training)
CT Projections (2/3/4 views)
FDK Reconstruction
Density-weighted Sampling (γ=1.0)
+Uniform (α=0.2)
50K Initial Gaussians
p(x) ∝ V_FDK(x)^γ, X = X_uniform ∪ X_weighted
Stage 2: GAR — Geometry-aware Refinement [Iters 2K-20K, every 500 iter]
Current Gaussians
KNN Proximity K=5, s(p)=mean(d)
s(p) < 0.015? → Candidate
Gradient < 0.0002?
Prune (max 2%/iter)
Refined Gaussians
s(p_i) = ¹/ₖ × Σ||p_i-p_j||₂, N_prune = min(N_cand, ⌊β·N_total⌋)
Training Iterations →
Stage 3: ADM — Adaptive Density Modulation [After 15K warmup]
Position x (x,y,z)
K-Planes F_xy+F_xz+F_yz (64×64×32)
MLP Decoder 3×128 units
Δρ = tanh(·)×r c = σ(·)
Zero-mean Modulation ρ_final=ρ_base+c·Δρ
Final Adjusted ρ